A Distributed Real-Time Intrusion Detection System on a Jetson Orin Nano Super Edge Cluster using Kafka and PySpark

Thai Nguyen Vu[1] Dung Van Bui[2] Nhut Tri Do[2] Van Du Nguyen[3]
SOICT 2026
[1]University of Transport and Communications, HCMC Campus [2]University of Information Technology, VNU-HCM
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Good morning. I will show what a second edge board actually buys you when you split an ML intrusion detector across two Jetson Orin Nano Super devices. Seven minutes, three results.

I. PROBLEM AND CONTRIBUTIONS

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Edge intrusion detection: three open gaps

Detectors now *fit* on single-board hardware — but a monolithic one classifies **every** flow and saturates an 8 GB ARM board.

What the literature does not tell us

1. Everything is measured on a **single device**.
2. Throughput, latency and **energy** are never reported together.
3. Offline scores come from **leaky protocols**.

Our question

How should inference be *placed* across a constrained two-node edge cluster?

Our answer

Split it: a cheap **autoencoder gate** on board #1, the heavy **PySpark classifier** on board #2, joined by a Kafka topic.

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Three gaps: single-device evaluation, energy never reported next to latency, and offline numbers taken from leaky pipelines. Which model to deploy is the other half of the question. We attack all three with one system.

Contributions

1. A **two-stage distributed IDS**: anomaly gate + Spark classifier, physically separated across two boards.
2. **Three deployment modes** for a two-node Jetson cluster — pipeline split, horizontal Kafka scaling, Spark standalone cluster.
3. An **end-to-end evaluation**: throughput, latency percentiles, per-node CPU/RAM *and* energy per verdict.
4. **Leakage-aware provenance** — the deployed model is exactly the one our companion study selected under a leakage-free protocol.

Contributions
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Four contributions. The fourth matters: the model we deploy is not a fresh model, it is the one that survived a strict offline audit.

II. SYSTEM AND EVALUATION SETUP

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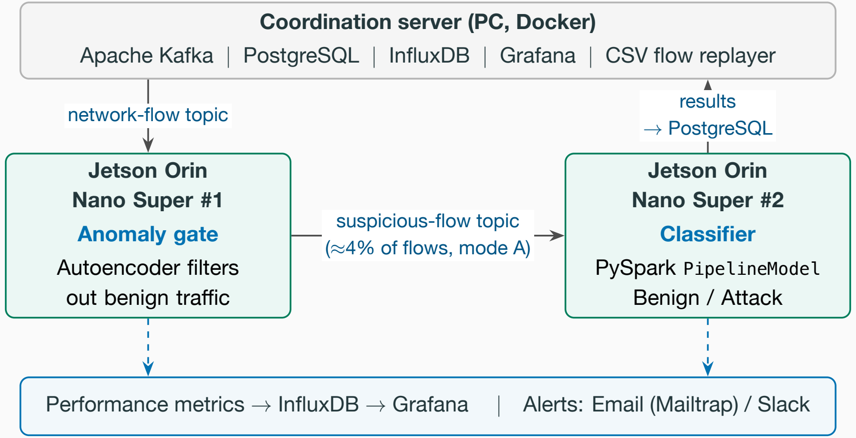
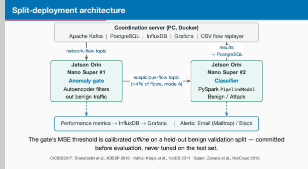
Split-deployment architecture



The gate's MSE threshold is calibrated *offline* on a held-out benign validation split — committed before evaluation, never tuned on the test set.

CICIDS2017: Sharafaldin et al., ICISSP 2018 · Kafka: Kreps et al., NetDB 2011 · Spark: Zaharia et al., HotCloud 2010.



Host runs the infrastructure. Board one gates, board two classifies. Only suspicious flows cross the Kafka topic between them. Everything is instrumented into Grafana.

Three deployment modes, one workload

Single node full pipeline in one process — the reference.

A: Pipeline split gate on #1, classifier on #2 (our design).

B: Horizontal both boards run the *full* role, sharing a Kafka consumer group.

C: Spark cluster host as master, both boards as workers.

Throughput is counted as **final verdicts per second** — a flow is decided at the gate or at the classifier — so forwarded flows are never double-counted.

Setup

$2\times$ Jetson Orin Nano Super (6-core A78AE, 8 GB, $\approx$67 TOPS)

PySpark 3.4.1, Kafka 3.5, Wi-Fi LAN

CICIDS2017 replay at 100 flows/s

$\geq$3 repetitions, warmup excluded

Energy via `tegrastats`, idle subtracted

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Energy via tegrastats, idle subtracted

Four configurations, identical workload. The verdict-per-second definition is what makes the modes comparable.

III. RESULTS AND CONCLUSION

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Result 1: splitting the pipeline buys 24× throughput

Mode	Throughput (flows/s)	p95 latency (ms)	Gate skip (%)	Energy/verdict (mJ)
Single node	2.60 ± 0.41	12.3 ± 3.0	(inline)	2490
A: Pipeline split	**62.5 ± 0.6**	**13.1 ± 2.6**	**95.8**	**178**
B: Horizontal	5.9 ± 1.4	21.6 ± 19.2	(inline)	2250
C: Spark cluster	90.0 ± 25.6	23.9 ± 17.8	97.9	119

- **A:** 24× **throughput at unchanged p95** — the gate absorbs 95.8% of benign traffic.
- **B stays Spark-bound** (≈2.3×): every node still classifies.
- C is fastest on average but swings by ±25.6 flows/s and needs the host master.
- Energy amortized over 24–35× more verdicts: ≈**2.5 J** → **0.12–0.18 J**.

Pipeline split is the recommended default — the highest throughput at the tightest run-to-run spread; cluster mode is for models that exceed single-node memory.

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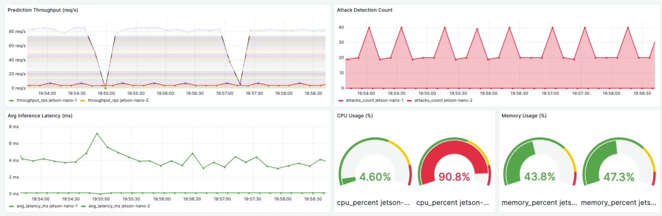
This is the core table. Twenty-four times, at the same p95. Mode B proves the gain comes from role separation, not from adding a board. Energy is total-board per verdict: the active power delta was below the tegrastats noise floor at this load.

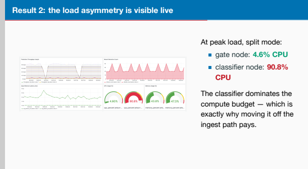
Result 2: the load asymmetry is visible live



At peak load, split mode:

- gate node: **4.6% CPU**
- classifier node: **90.8% CPU**

The classifier dominates the compute budget — which is exactly why moving it off the ingest path pays.



Live Grafana at peak load. Under five versus ninety-one percent. The asymmetry is the whole argument for the split. Add: the gate is a tunable filter, not a fixed point — backup slide has the operating curve.

Result 3: the price of one unified stack

We serve with Spark for *consistency with distributed training*. What does that cost? Same forest, same 29 deployed features, same board:

Engine	Throughput (flows/s)	p95 (ms)	RAM (%)	Energy/inf. (mJ, active)
PySpark (deployed)	22.6	532	24.3	71.6
scikit-learn	204.4	49.8	13.3	2.91
ONNX Runtime	**7870.8**	**1.1**	15.7	**0.06**

ONNX Runtime sustains 348× the deployed PySpark throughput. We report this as a *target for the next iteration* — export to ONNX/TensorRT for serving; keep Spark for training.

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The JVM costs us three orders of magnitude, and that sets the roadmap.

Conclusion

What we showed

- A reproducible distributed edge IDS: Kafka + PySpark on a Jetson Orin Nano Super cluster.
- Splitting gate and classifier across boards: **2.6 → 62.5 flows/s** at p95 ≈13 ms, gate skipping 95.8%.
- Energy per verdict: **2.5 J → 0.18 J**.

Limitations

- Two-node lab cluster over Wi-Fi; CICIDS2017 replay, not live traffic.
- The ≈67-TOPS GPU is unused by the CPU-only pipeline.
- Evasive traffic untested.

Next

- TensorRT/ONNX in the streaming path.
- Live traffic; federated retraining under drift.

Thank you — questions?

Recap the three numbers: twenty-four times, thirteen milliseconds, ninety-six percent skipped. Then stop talking.

Key references

[1] Sharafaldin, I., Lashkari, A. H. & Ghorbani, A. A. *Toward generating a new intrusion detection dataset and intrusion traffic characterization*. ICISSP 2018. **CICIDS2017**

[2] Shi, W. et al. *Edge computing: Vision and challenges*. IEEE Internet of Things Journal 3(5), 2016. **edge setting**

[3] Chandola, V., Banerjee, A. & Kumar, V. *Anomaly detection: A survey*. ACM Computing Surveys 41(3), 2009. **anomaly gate**

[4] Lundberg, S. M. & Lee, S.-I. *A unified approach to interpreting model predictions*. NeurIPS 2017. **SHAP Top-30**

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[6] Kreps, J., Narkhede, N. & Rao, J. *Kafka: a Distributed Messaging System for Log Processing*. NetDB 2011. **Kafka**

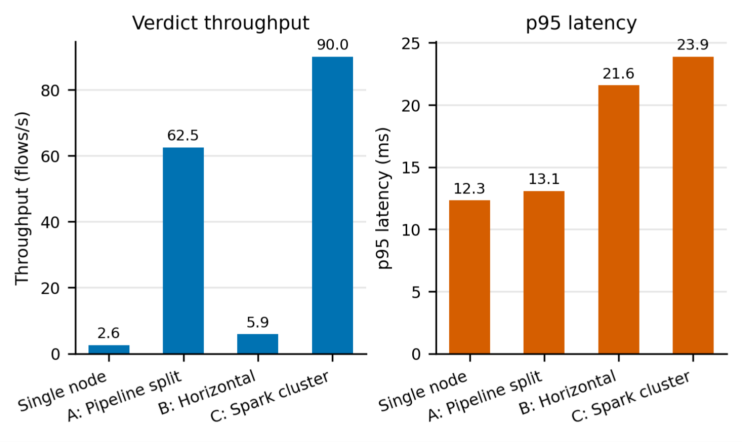
[7] Mohan Kiran Kumar, K. et al. *Distributed Intrusion Detection System using Kafka and Spark Streaming*. ICVADV 2025. **closest system**

[8] Arp, D. et al. *Dos and Don'ts of Machine Learning in Computer Security*. USENIX Security 2022. **leakage-aware protocol**

Not presented — shown while taking questions, so sources are on screen if the audience asks. Full list is in the paper.

Backup slides

Backup: deployment-mode comparison chart



Verdict throughput (left) and p95 latency (right)

2026-09-04

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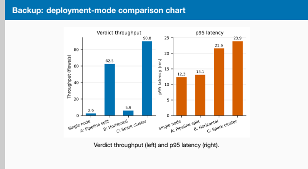
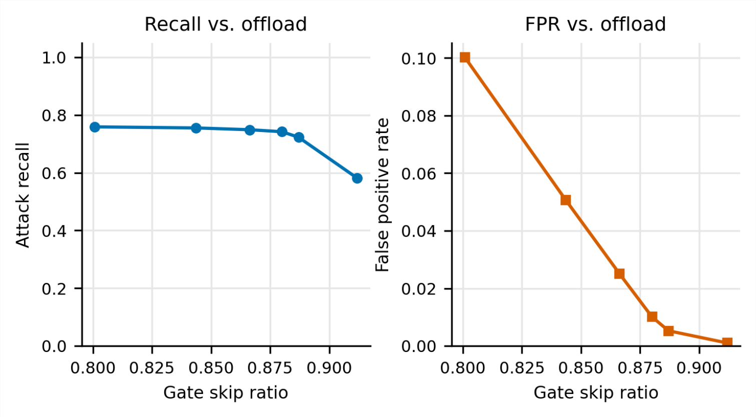
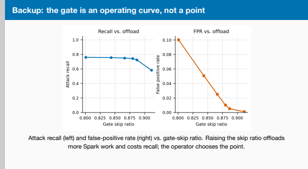


Backup: the gate is an operating curve, not a point



Attack recall (left) and false-positive rate (right) vs. gate-skip ratio. Raising the skip ratio offloads more Spark work and costs recall; the operator chooses the point.



Backup: which model, and why

Configuration	F1	Features	Size (MB)	Edge rationale
Baseline + XGBoost	0.9971	77	2.32	Reference accuracy
SHAP Top-30 + XGBoost	0.9970	30	0.003†	Explainable; deployed feature set
RF Top-30 + XGBoost	0.9922	30	2.52	Embedded ranking
PCA k=40 + XGBoost	0.9939	40	3.82	Unsupervised reduction

We deploy **SHAP Top-30 with Random Forest** (offline F1 0.9955): a native Spark ML estimator, so the exported `PipelineModel` loads on ARM64 without the native libraries Spark's XGBoost integration needs. The input vector has **29** features — `destination_port` is in the SHAP ranking but excluded as label-leaking.

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Backup: full monitoring dashboard

Per-node throughput, detected attacks, inference latency, CPU/RAM for both boards, PostgreSQL-backed alerts.

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Backup: measurement protocol

- Host-side orchestration starts pipeline roles on the Jetsons over SSH, drives the load, collects per-node logs.
- $\geq$3 repetitions per configuration; warmup traffic *excluded* from the measured window (no JVM/JIT inflation of early batches).
- Metric queries aligned to the exact load window, not a trailing range.
- Latency percentiles per node over *raw per-flow* samples; cluster p95 is conservatively the *worst node's* p95.
- End-to-end send-to-verdict latency from producer timestamps under NTP-synchronized clocks (excludes DB-write cost).
- Energy: `tegrastats`, 30 s idle baseline then load window; two-node modes sum both boards.

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